

FEDERATED LEARNING PRIVACY AND SECURITY ENHANCEMENT

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Abstract—Federated Learning (FL) has emerged as a promising alternative to centralized machine learning by enabling collaborative model training without direct data sharing. With recent advances in FL on feature analysis, aggregation algorithms, privacy, and security, FL improves data locality and lowers raw data exposure compared to centralized learning. However, it introduces new methods of attacks such as model poisoning, inference/membership attacks and important tradeoffs among communication cost, robustness, and privacy guarantees. We compare classical techniques such as federated averaging with robust aggregation strategies including Trimmed Mean, Krum, Bulyan, and Robust Federated Aggregation, highlighting trade-offs in computational efficiency, scalability, and resilience against compromised clients. Our analysis shows how FL improves data privacy through secure aggregation and differential privacy, and how it deals with threats such as poisoning and inference attacks. A survey of recent patents shows developing interest in communication efficiency, adaptive aggregation and privacy solutions. Overall, FL demonstrates strong potential for privacy-first machine learning, but faces challenges in robustness, communication overhead, and data heterogeneity that define areas of interest for future research.

I. INTRODUCTION

A. FL Feature Analysis vs. Centralized Learning

Traditional centralized machine learning systems aggregate raw data from multiple clients into a single location/datacenter for training. While this approach simplifies optimization and model management, it raises critical challenges concerning privacy, security, and scalability when applied across diverse and distributed data sources. Federated Learning (FL) addresses these issues by keeping data local to client devices and transmitting only model updates to a central server for aggregation, so important data stays on your device.

Federated Averaging (FedAvg) was one of the algorithms introduced early on that served as the foundational aggregation algorithm, providing a balance between communication efficiency and scalability. However, FedAvg and related methods face limitations in heterogeneous environments and are vulnerable to manipulation. This has led to the development of alternative aggregation techniques such as Trimmed Mean, Krum, Bulyan, and Robust Federated Aggregation (RFA), which emphasizes robustness to data poisoning and Byzantine attacks. Recent advancements also incorporate adaptive optimizers and heterogeneity aware methods to improve convergence and fairness among clients.

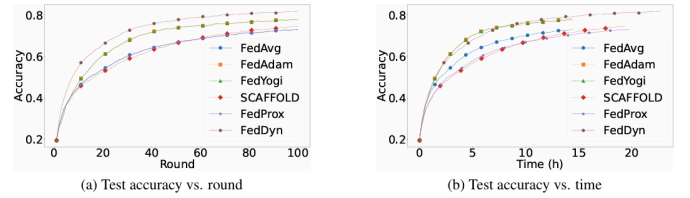


Figure 1. “Test accuracy for CIFAR-10 dataset with Dir(0.3) and 100 clients. We compare the accuracy across algorithms by choosing round or time on the x-axis.” (Baumgart et al., 2024)

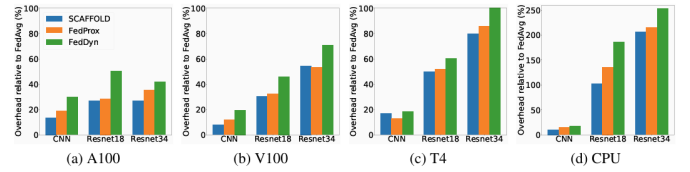


Figure 2. “The relative runtime overhead of algorithms compared to FedAvg’s runtime with CNN (798K parameters), ResNet18 (11.7M), and ResNet34(21.8M).” (Baumgart et al., 2024)

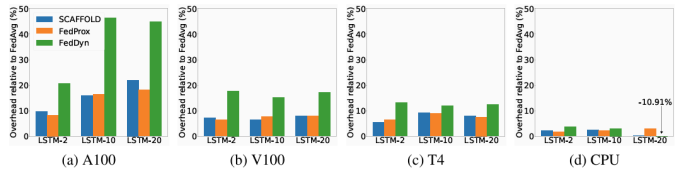


Figure 3. “The relative runtime overhead of algorithms compared to FedAvg’s runtime with LSTM-2 (134K parameters), LSTM-10 (780K), and LSTM20 (1.59M).” (Baumgart et al., 2024)

B. Privacy and Security in Federated Learning

Although FL reduces the risk of centralized data exposure, it is not immune to privacy and security threats. Client updates can leak sensitive information through gradient inversion or membership inference attacks. To mitigate these risks, research has introduced several defenses such as secure aggregation protocols, which ensure that only aggregated updates are visible to the server; differential privacy (DP), which adds calibrated noise to protect against inference; and trusted execution environments (TEEs), which safeguard computations at the hardware level.

Despite these protections, FL systems remain vulnerable to model poisoning and backdoor attacks, where malicious clients deliberately inject manipulated updates to degrade or bias the global model. Case studies in healthcare and finance have demonstrated the severity of such risks, prompting the integration of anomaly detection and robust aggregation strategies into modern FL frameworks.

C. Challenges and Future Directions

While FL has demonstrated strong potential in privacy preserving machine learning, several barriers prevent its adoption on a larger scale. Communication efficiency remains a bottleneck, as frequent exchanges of model updates can overwhelm network resources. Model heterogeneity and non-IID data distributions pose further challenges, leading to slower convergence and fairness concerns among clients. Additionally, the scarcity of labeled data in decentralized environments complicates supervised training on larger scales.

Future research directions include the design of more efficient communication protocols, integration of meta-learning and metacognitive control for adaptive model updates, and novel strategies to address data heterogeneity and personalization. Industry interest is also reflected in recent patents that emphasize adaptive aggregation, communication efficient protocols, and secure system level deployments, signaling a pathway toward more robust and large scale federated learning systems.

II. DATA SET AND PREPARATION

A. Data Selection

For this project, the Medical Segmentation Decathlon (MSD) dataset, a large-scale benchmark collection designed to advance research in medical image analysis, was selected. The dataset encompasses multiple imaging modalities (MRI, CT) across diverse anatomical regions such as the brain, lung, heart, pancreas, and prostate. Each sub-dataset consists of volumetric scans paired with expert-provided segmentation masks, enabling supervised training for medical image segmentation tasks. The MSD dataset was chosen for its clinical relevance and inherent heterogeneity, making it a strong candidate for evaluating federated learning (FL) in healthcare scenarios where data privacy is necessary.

B. Dataset Preprocessing

Since MSD is not structured for FL, the following data preparation strategy was implemented to simulate distributed, privacy-preserving training across multiple clients.

- Data was cleaned by removing duplicate or corrupted scans to ensure consistency and basic denoising filters were applied to reduce imaging artifacts that could affect segmentation accuracy.
- Data was normalized to a fixed range $[0,1]$ across all scans to reduce variability across imaging devices and facilitate faster convergence.
- To improve model generalization, training data was augmented with random affine transformations such as rotations, translations, zooming, and elastic deformations. Additionally, Horizontal and vertical flips were selectively applied where possible.

C. Dataset Preparation

To prepare the data for FL, the dataset was partitioned into virtual clients representing different institutions. To reflect real world federated settings, we adopted a non-IID distribution, each client was assigned scans from only a subset of anatomical sites, resulting in label imbalance and varied feature distributions. Clients maintained separate local datasets while participating in global training through federated aggregation. Segmentation masks were verified for format consistency and aligned with input scans. Labels were standardized across clients to ensure uniformity in class definitions. Following project guidelines, we introduced controlled label corruption for a subset of clients to simulate adversarial or noisy data contributions. This allows us to assess the robustness of FL aggregation strategies under practical conditions. Through these preparation steps, the MSD dataset was adapted from its original centralized design into a federated learning environment. This setup more realistically mirrors clinical use cases, where sensitive medical data remains siloed at local institutions, but collaborative model training is still desired.

III. MODEL SELECTION AND DISTRIBUTION

A. Model Selection and Reasoning

The pre-trained U-Net will be adapted to the MSD dataset by modifying its decoder and output layers to align with the segmentation tasks. The encoder backbone (EfficientNet due to hardware limitations) will be initialized with ImageNet pre-trained weights. The decoder layers will be initialized randomly using Xavier initialization. The final output layer will be adjusted to produce segmentation masks with the appropriate number of classes for the selected MSD task (e.g., binary tumor segmentation or multi-class organ delineation). This hybrid initialization strategy leverages pre-trained knowledge while retaining flexibility for task-specific adaptation.

The choice of U-Net with a pre-trained backbone balances model complexity and communication efficiency, making it feasible for FL settings where clients may have limited computational resources. Furthermore, initializing from a pre-trained model improves the generalization ability of the model across heterogeneous client data, which is essential in medical applications where data is highly non-IID.

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